

REPORT ON

STOCK PRICE PREDICTION USING MACHINE LEARNING

1. INTRODUCTION

Stock price prediction is a practical application of data analysis and machine learning. This project analyzes historical stock variables and uses regression techniques to estimate the next-day closing price.

2. OBJECTIVES

- Perform EDA and understand stock price trends.
- Analyze Open, High, Low, Close, and Volume.
- Handle missing values and examine outliers.
- Create 7-day and 30-day moving averages, daily returns, and volatility.
- Scale features where required.
- Build and compare Linear and Polynomial Regression.
- Evaluate MAE, MSE, and R^2 .
- Perform time-series cross-validation.
- Create trend, moving-average, correlation, and prediction visualizations.

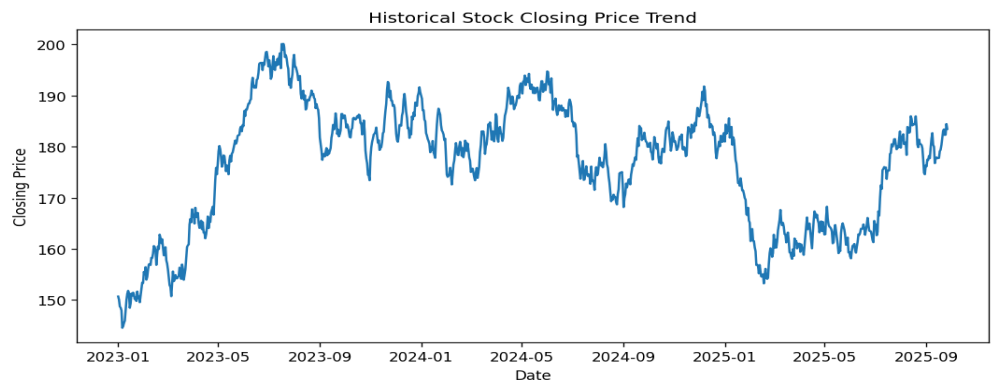
3. DATASET DESCRIPTION

The dataset contains 1,000 observations before preprocessing and includes daily market variables plus Target_Next_Close. After rolling-feature preprocessing, 971 observations were used for modelling.

Feature	Description
Date	Trading date
Open	Opening stock price
High	Highest price during the day
Low	Lowest price during the day
Close	Closing stock price
Volume	Number of shares traded
MA_5	5-day moving average
MA_20	20-day moving average
MA_7	7-day moving average
MA_30	30-day moving average
Daily_Return	Daily percentage return
Volatility_7	7-day rolling volatility
Target_Next_Close	Next-day closing price

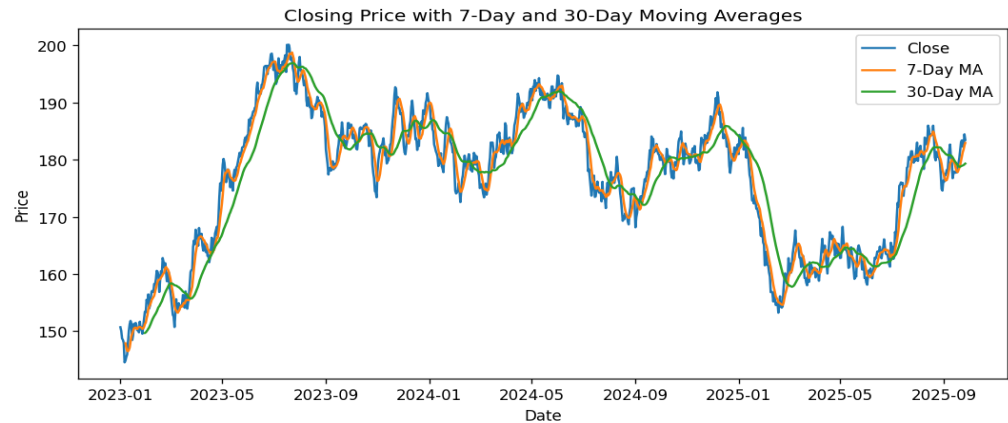
4. EXPLORATORY DATA ANALYSIS

4.1 STOCK CLOSING PRICE TREND



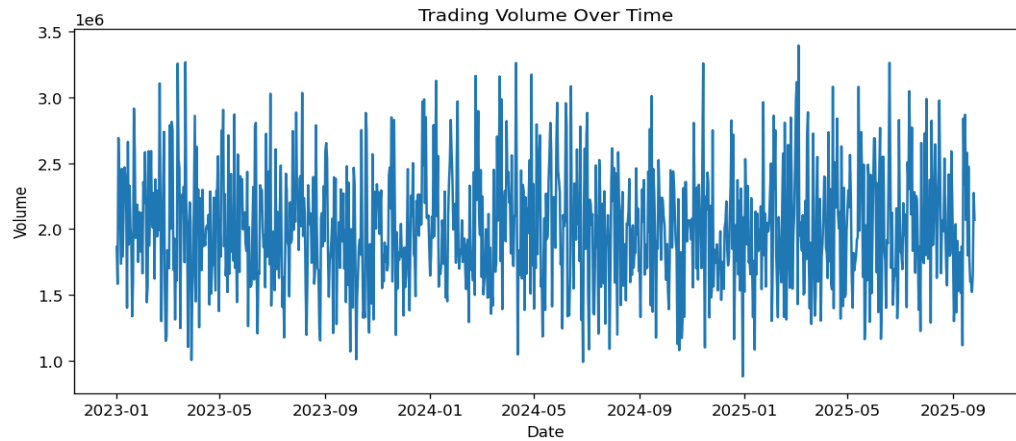
Observation: The closing price shows fluctuations over the selected period with upward and downward movements.

4.2 MOVING AVERAGES



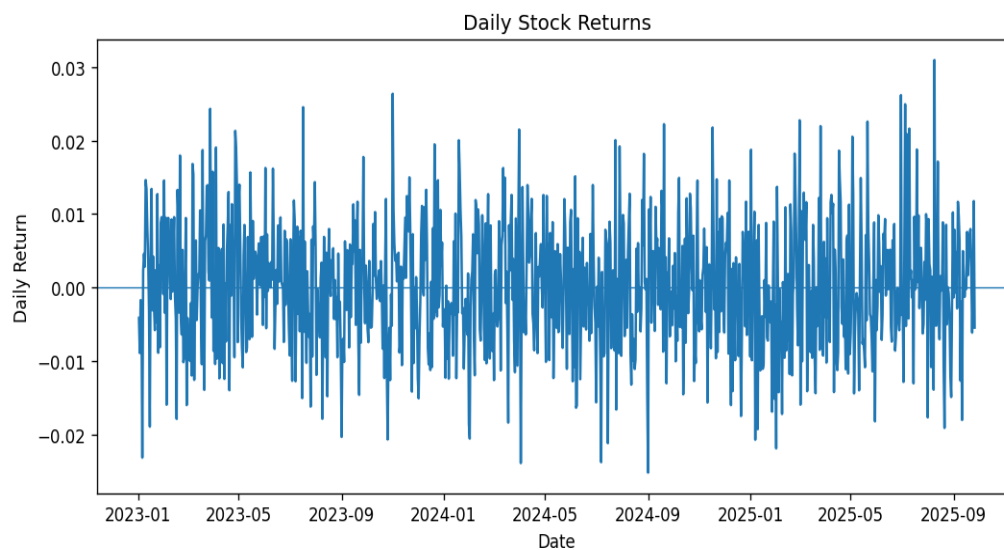
Observation: The 7-day moving average captures short-term movement, while the 30-day moving average provides a smoother longer-term trend.

4.3 TRADING VOLUME OVER TIME



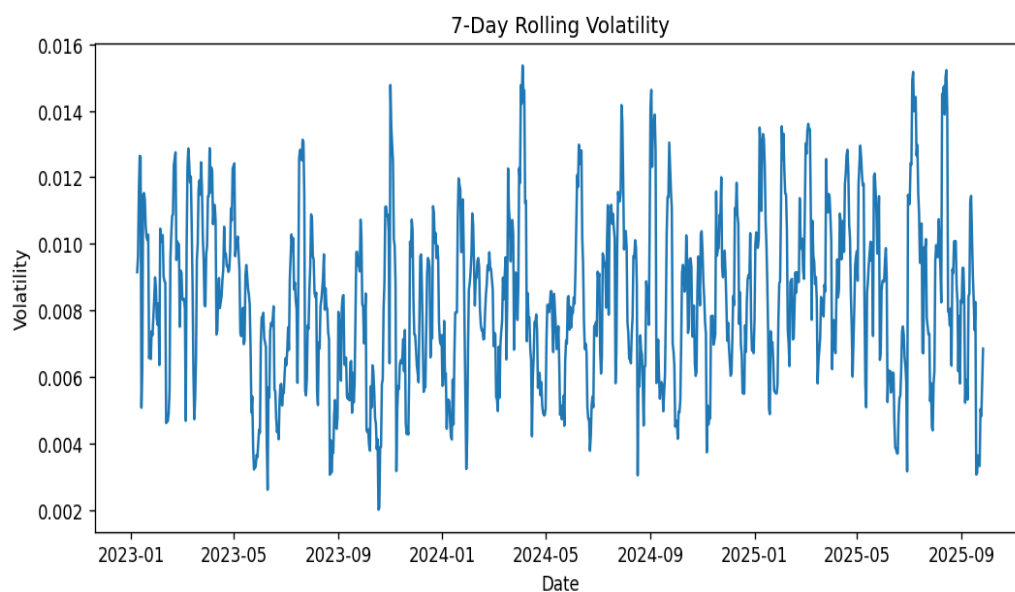
Observation: Trading volume varies across the sample and reflects changing market activity.

4.4 DAILY STOCK RETURNS



Observation: Returns are generally near zero, with larger positive or negative values indicating stronger daily movements.

4.5 DAY ROLLING VOLATILITY

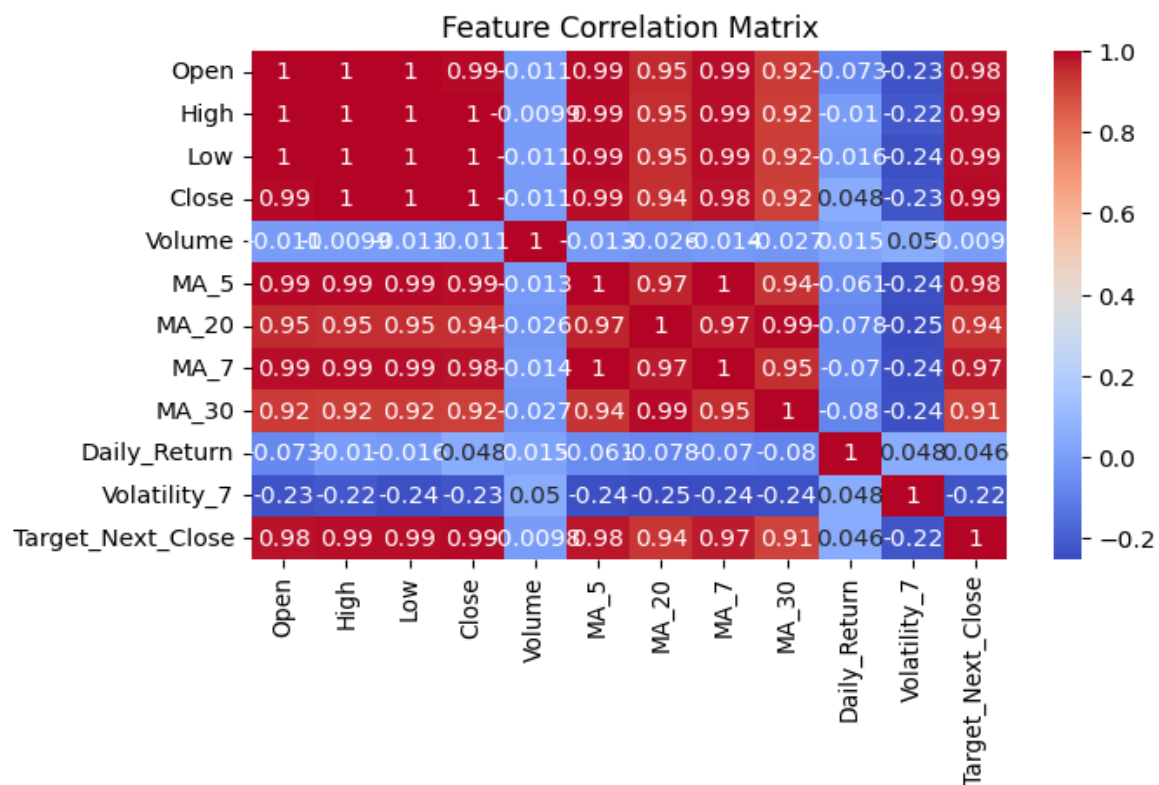


Observation: Volatility changes over time and identifies periods of relatively higher or lower short-term uncertainty.

5. DATA PREPROCESSING

The Date column was converted to date time and sorted chronologically. The original dataset contained no missing values. Initial rows without sufficient history for rolling features were removed. Potential extreme observations were considered in market context and were not automatically deleted because they may represent genuine price movements.

6. CORRELATION ANALYSIS



Observation: Close has a very strong positive correlation with Target_Next_Close (approximately 0.991). Open, High, Low, and moving averages also show strong positive relationships, while Volume is weakly related.

7. MODEL DEVELOPMENT

The predictors were Open, High, Low, Close, Volume, MA_5, MA_20, MA_7, MA_30, Daily_Return, and Volatility_7. Target_Next_Close was the response variable.

7.1 TRAIN-TEST SPLIT

An 80:20 chronological split was used without shuffling to respect the time-dependent nature of stock data.

7.2 SCALING

Standard Scaler was used inside Linear and Polynomial Regression pipelines.

7.3 LINEAR REGRESSION

Linear Regression was used as the baseline model.

7.4 POLYNOMIAL REGRESSION

Degree-2 Polynomial Regression was used to test nonlinear feature relationships.

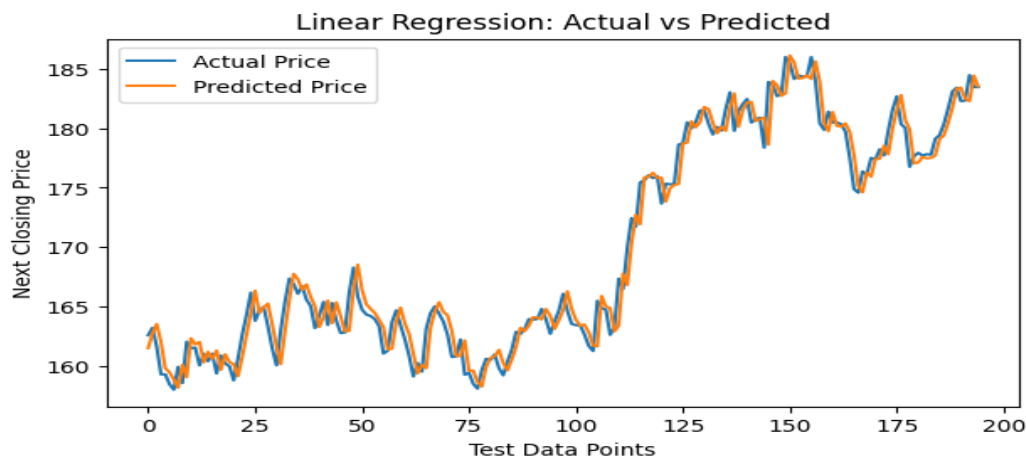
7.5 DECISION TREE

A Decision Tree Regressor with maximum depth 5 was used as an optional nonlinear benchmark.

8. MODEL EVALUATION

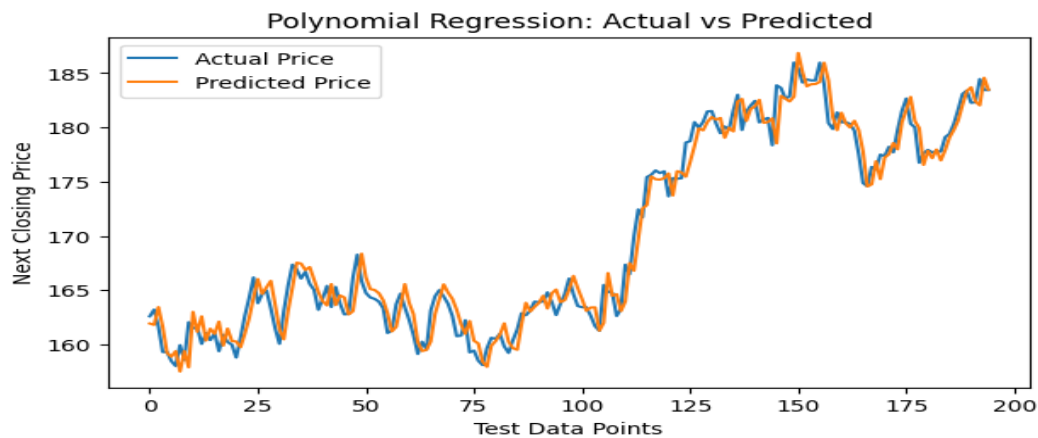
Model	MAE	MSE	R ²
Linear Regression	1.205	2.338	0.971
Polynomial Regression	1.272	2.524	0.968
Decision Tree	1.479	3.686	0.954

8.1 ACTUAL VS PREDICTED: LINEAR REGRESSION



Observation: Predicted values are close to actual next-day closing prices, indicating a strong test-set fit.

8.2 ACTUAL VS PREDICTED: POLYNOMIAL REGRESSION



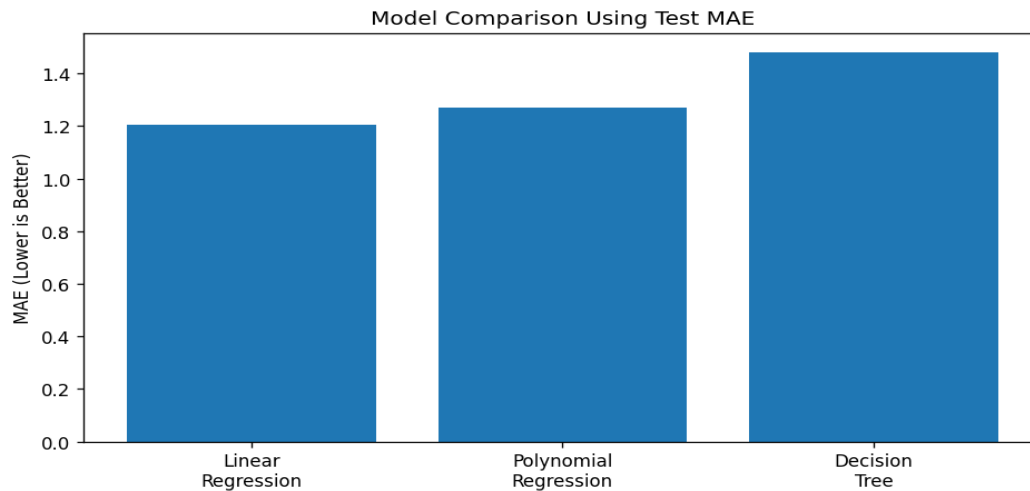
Observation: Polynomial Regression follows actual values closely but has slightly higher errors than Linear Regression

8.3 CROSS-VALIDATION

Model	CV MAE	CV MSE	CV R ²
Linear Regression	1.299	2.617	0.940
Polynomial Regression	1.581	4.348	0.888
Decision Tree	1.924	7.267	0.802

Five-fold Time Series Split cross-validation was used. Linear Regression achieved the lowest average CV MAE and CV MSE and the highest CV R^2 .

8.4 MODEL COMPARISON USING TEST MAE



Observation: Linear Regression has the lowest MAE; lower MAE means smaller average absolute prediction error.

9. KEY OBSERVATIONS

1. The stock closing price shows noticeable fluctuations over time.
2. The 7-day moving average captures short-term price behavior, while the 30-day moving average provides a smoother long-term trend.
3. Close, High, Low, and Open prices have strong relationships with the next-day closing price.
4. Volume has a relatively weak relationship with the prediction target.
5. Linear Regression achieved the best performance among the tested models.
6. Polynomial Regression did not improve the prediction performance compared with Linear Regression.
7. Decision Tree Regression produced comparatively higher errors.
8. Cross-validation results also support Linear Regression as the most stable model for this dataset.

10. RECOMMENDATIONS

Linear Regression is recommended for this dataset because it provides the lowest errors and highest R^2 . Future work could add market indices, sector indicators, macroeconomic variables, sentiment, and additional technical indicators.

19. LIMITATIONS

The model is based mainly on historical market variables and cannot capture every external factor affecting stock prices. News, economic events, sentiment, and regime changes may reduce future predictive performance. High historical R^2 does not guarantee future trading profits.

20. CONCLUSION

This project demonstrated a complete stock-price prediction workflow using EDA, feature engineering, regression modelling, evaluation, visualization, and time-series cross-validation. Linear Regression was the best model on the test set with MAE = 1.205, MSE = 2.338, and R^2 = 0.971. Cross-validation also favored Linear Regression, so it was selected as the preferred model for this dataset.